

# **Module 2**

## Filesystem, Permissions & Environment

# Learning objectives

Read ls -l output (types, permissions, owner)

Use chmod to make scripts executable safely

Explain HOME, PATH, and which

Fix common permission denied and command not found errors

# Prerequisites

Module 1 complete (cd, ls, pwd, basic files)

# Learning path

## Learning Path

Diagram: Learning Path  
(render with mmdc when available)  
flowchart LR  
A[File types] --> B[Permissions]  
B --> C[Ownership]  
C --> D[Environment PATH]  
D --> E[Dotfiles]  
E --> F[Module 3]

# Course context

Simulators and tools depend on PATH

Scripts in projects need chmod +x

Dotfiles like .gitignore appear in every repo

# Core topics

# Self-check

```
$ ./scripts/module2.sh --check
```

Terminal: module\_check

```

[0;36m=====
[0;36mModule 2: Environment self-check
[0;36m=====

[0;32m[OK][0m ls available
[0;32m[OK][0m chmod available
[0;32m[OK][0m which available
[0;32m[OK][0m readlink available
[0;32m[OK][0m HOME set: /home/yongfu
[0;32m[OK][0m PATH set
[0;32m[OK][0m module2 directory exists: /mnt/d/proj/universal-verification-methodology/learn_unix_
[0;32m[OK][0m file_types example exists
[0;32m[OK][0m permissions/hello.sh exists
[0;32m[OK][0m file_types symlink (my_symlink) exists
[0;32m[OK][0m environment/inspect_env.sh exists
[0;32m[OK][0m path_and_script/local_cmd.sh exists and is executable

[0;32m[OK][0m All required checks passed.
```

# Make a script executable

```
chmod u+x hello.sh  
./hello.sh
```



# ls -l and permissions

```
$ cd module2/examples/permissions && ls -l hello.sh config.txt
```

Terminal: permissions\_ls

```
-rwxrwxrwx 1 yongfu yongfu 145 May 28 12:03 config.txt  
-rwxrwxrwx 1 yongfu yongfu 150 May 28 12:03 hello.sh
```

# Finding commands on PATH

```
$ which bash && which git
```

```
Terminal: which_tools
```

```
/usr/bin/bash  
/usr/bin/git
```

# Practice

# Exercise scaffold

```
./scripts/module2.sh --scaffold
```

# Exercises

chmod on ~/unix\_practice/hello.sh

echo \$PATH and which for git

ls -la to see dotfiles in examples/dotfiles

# Summary & next steps

File types and rwx permissions

PATH and environment

Next: Module 3 — processes & pipes

Next: Module 3: Processes, Pipes & Redirection